

BIM & Grant Funding

Building Information Modeling · USDA DLT Grant · Property Grant Strategy

SECTION 1 — What is BIM?

Building Information Modeling (BIM) is a digital process for creating and managing all information about a building — from design and construction through to operation and demolition. It produces a rich 3D model containing not just geometry but data: materials, costs, schedules, energy performance, and more.

Key Dimensions of BIM

Dimension	What It Covers
3D BIM	3D geometry — walls, roofs, doors, windows, structure
4D BIM	Time / construction schedule linked to the 3D model
5D BIM	Cost estimation and quantity take-offs
6D BIM	Sustainability & energy analysis
7D BIM	Facility management and operations data

Core Benefits of BIM

- Reduces design errors by up to **70%** through early clash detection
- Cuts project costs by **10–30%** and timelines by **15–25%**
- Enables accurate cost estimation before construction begins
- Improves collaboration between architects, engineers, and contractors
- Supports energy modeling and sustainability compliance
- Creates a digital twin for long-term facility management
- Strengthens grant applications with precise, verifiable building data

Common BIM Software

Software	Primary Use	Provider
Autodesk Revit	Full BIM authoring (architecture, MEP, structure)	Autodesk
AutoCAD	2D drafting + BIM integration	Autodesk
Navisworks	Clash detection & 4D simulation	Autodesk
ArchiCAD	Architectural BIM modeling	Graphisoft

SketchUp	Conceptual 3D modeling	Trimble
Blender (bpy)	3D visualization & scripted building generation	Open Source

SECTION 2 — How BIM Strengthens Grant Applications

Grant agencies require detailed, credible project documentation. A BIM model provides precise measurements, cost data, energy performance, and visual evidence that a project is well-planned and fundable.

How to Use a BIM Model in a Grant Application

- **Accurate Cost Estimates:** BIM's 5D capability generates quantity take-offs and material costs — directly usable in grant budget sections.
- **Site & Building Documentation:** Export floor plans, elevations, and 3D renders to attach as project site documentation.
- **Energy & Sustainability Data:** 6D BIM energy analysis supports grants prioritizing climate resilience and environmental justice.
- **Accessibility Compliance:** BIM can verify ADA/Section 504 compliance — required for federal grants.
- **Scope of Work Narrative:** BIM model data feeds directly into the Scope of Work section of applications.
- **Matching Funds Valuation:** In-kind contributions (existing equipment, building improvements) can be valued using BIM data.

SECTION 3 — USDA DLT Grant (Distance Learning & Telemedicine)

Program Overview

Agency	USDA Rural Utilities Service (RUS), Rural Development
Program	Distance Learning & Telemedicine (DLT) Grant
Authority	7 U.S.C. 950aaa 7 CFR Part 1734
FY 2025 Funds	~\$40 million available
Award Range	\$50,000 minimum — \$1,000,000 maximum
Match Required	15% of grant amount (cash or in-kind)
Performance Period	3 years from date funds are released
Submit Via	www.grants.gov (electronic only)

Purpose of the DLT Grant

The DLT program funds telecommunications equipment and technology that enables **distance learning** and **telemedicine** services in rural areas. It connects rural residents to education, training, and healthcare resources that are otherwise unavailable or limited.

Who Can Apply

- School districts and educational institutions
- Healthcare providers (hospitals, clinics, rural health centers)
- Libraries and community learning centers
- State and local government agencies
- Federally Recognized Tribes
- Non-profit organizations providing education or healthcare
- Faith-based organizations (on equal basis with others)

What the Grant Funds

- Telecommunications equipment (video conferencing systems, interactive displays)
- Computer hardware and networking equipment at hub and end-user sites
- Video conferencing platform licenses (up to 10% of grant, 3-year limit)
- Installation and technical setup costs
- Construction costs related to DLT infrastructure (
- Telemedicine diagnostic equipment and distance learning technology

Scoring Criteria

Criterion	Max Points	How to Score High
Rurality	40 pts	Smaller population = higher score; use 2020 Census data
Economic Need	30 pts	Higher county poverty % = higher score (SAIPE data)
Service Need & Benefits	30 pts	Document local healthcare/education gaps clearly
Special Consideration	10 pts	Distressed Communities Index >80, or SVI ≥0.75

SECTION 4 — Our Property: Ravenswood, West Virginia

The property is located in **Ravenswood, Jackson County, West Virginia** — a small rural community that strongly aligns with USDA DLT grant eligibility and scoring criteria.

Why This Property is a Strong Grant Candidate

- Ravenswood population is under 4,000 — qualifies for **maximum Rurality points (40 pts)**
- Jackson County, WV poverty rate is above the national average — strong **Economic Need score (30 pts)**
- West Virginia ranks among the most underserved states for healthcare access and broadband

- Limited local access to medical specialists and higher education — directly matches DLT goals
- The region may qualify for **Distressed Communities Index** special consideration points (10 pts)
- WV is eligible for USDA Rural Development programs across all community facility categories

Grant Fit Summary: A building in Ravenswood, WV used for telemedicine or distance learning could score near the maximum under the DLT program due to its rural location, economic need, and lack of existing services — making it a highly competitive application.

SECTION 5 — Adding the Building to the DLT Grant

The Ravenswood property can serve as a **hub site** or **end-user site**. Follow these steps to include it in a DLT grant application.

Step 1 — Confirm Rural Eligibility

Use the DLT mapping tool at rd.usda.gov to verify the Ravenswood address. Jackson County, WV is expected to qualify.

Step 2 — Define the Building's Role

Designate as Hub Site or End-User Site. Community buildings typically serve as end-user sites.

Step 3 — Identify the Service

Choose the DLT service: telemedicine clinic room, distance learning classroom, or community health/education center.

Step 4 — Document with BIM

Create a BIM model of the property using Revit, ArchiCAD, or Blender. Export floor plans as site documentation.

Step 5 — Prepare the Site Worksheet

Complete USDA's site worksheet: site name, address, GPS coordinates, population served, and current services available.

Step 6 — Budget the Equipment

List all DLT equipment needed and obtain 3 independent vendor quotes.

Step 7 — Secure 15% Match

Arrange matching funds = 15% of grant requested via local government contribution, in-kind donation, or cash match.

Step 8 — Write the Narrative

Document Jackson County's healthcare/education gaps, how the building addresses them, community involvement, and post-grant sustainability.

SECTION 6 — Other Grants for the Property

Beyond the DLT grant, several other federal and USDA programs can fund rural buildings and community facilities in West Virginia.

Grant Program	Funder	What It Funds	Best Fit
Community Facilities Direct Loan & Grant	USDA Rural Development	Construction, renovation, or purchase of community facilities	Clinic, school, or public community center
Community Facilities Guaranteed Loan	USDA Rural Development	Loan guarantees for essential community facilities	Larger construction or renovation projects
Rural Economic Development Grant	USDA RUS	Job creation & rural business development	Building hosting rural business or workforce training
Rural Community Development Initiative	USDA Rural Development	Capacity building for low-income rural communities	Non-profit managing building for community services
HUD Community Development Block Grant	HUD	Housing, infrastructure, community facilities	Building in low-income census tract
E-Rate Program	FCC / USAC	Broadband & telecom for schools & libraries	Building used as a school or library site